

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau

Phase 5 – Project Development Phase

## 1. Data Collection & Extraction from Database

The cosmetics dataset is collected containing key attributes: brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, Sensitive). The dataset is connected to Tableau Desktop/Public via Data → Connect → Text File / Excel, allowing Tableau to read and process the data directly for visualization.

## 2. Data Preparation

Steps taken to prepare the data for visualization:

- Checked and removed nulls/duplicate records across brand, price, and rank fields.
- Split/pivoted the skin-type column where multiple skin types were listed in a single row, so each type could be analyzed independently.
- Created calculated field 'Price Tier' (Budget / Mid / Premium) using an IF/CASE formula on the price field.
- Created calculated field 'Rank Category' (Top Rated / Low Rated) based on the rank field.
- Verified data types (numeric fields for price and rank, string fields for brand and category).

## 3. Data Visualizations

### Number of Unique Visualizations: 6

- Bar Chart — Product count by category and by brand.
- Box Plot / Histogram — Price distribution across categories.
- Heat Map — Brand vs. skin-type coverage.
- Scatter Plot — Price vs. Rank (value-for-money analysis).
- Treemap — Market share by brand.
- Pie / Bar Chart — Skin-type suitability breakdown.

## 4. Dashboard

The dashboard combines 3–4 of the above visualizations into a single interactive view, with filters for Brand, Category, Skin Type, and Price Range. The layout uses Tableau's device-specific 'Fit Width' design settings to ensure it remains responsive across desktop, tablet, and mobile screens.

## 5. Story

Number of Scenes: 3 — one per business scenario:

- Scene 1: Monitoring Consumer Preferences — highlights declining interest trends with supporting visuals.
- Scene 2: Addressing Product Concerns — shows negative-review/safety patterns and affected products.
- Scene 3: Predictive Analysis and Product Innovation — presents historical trend patterns to guide future decisions.

## 6. Performance Testing

| Metric                       | Detail                                                                 |
|------------------------------|------------------------------------------------------------------------|
| Amount of Data Loaded        | Full cosmetics dataset loaded and validated for completeness           |
| Utilization of Data Filters  | Brand, Category, Skin Type, and Price Range filters applied and tested |
| No. of Calculation Fields    | 2+ (Price Tier, Rank Category)                                         |
| No. of Visualizations/Graphs | 6 unique visualizations across the dashboard and story                 |

## 7. User Acceptance Testing (UAT)

Once built, the dashboard and story are shared with a sample stakeholder (e.g., a peer acting as a brand manager) to confirm the visuals are easy to interpret, filters behave as expected, and the story clearly communicates the three business scenarios. Feedback is collected and any minor adjustments (labels, colors, filter placement) are made before finalizing.

## 8. Web Integration

The published Tableau dashboard and story are embedded into a Flask web application using Tableau's embed code (iframe or Tableau JavaScript API) inside an HTML template, giving stakeholders a single web-based access point.

This completes Phase 5 (Project Development), covering: Data Collection, Data Preparation, Data Visualizations, Dashboard, Story, Performance Testing, User Acceptance Testing, and Web Integration.